

EXPERTS LIVE GERMANY 2026

Hybrid by Design

Wie Azure Arc lokale Server wieder sexy macht

Haiko Hertes

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01

Intro

Vielen Dank an unsere Sponsoren!

Diamond



Platinum



Gold



Silber





- Seit 2019 bei SoftwareOne
- Principal Consultant & Architect
- Azure Consulting Team
- Microsoft MVP, YouTuber, Blogger, Conference Speaker
- Familienvater, Offizier d.R., Holzwurm



www.hertes.net



about.me/haiko.hertes



twitter.com/HHertes



youtube.com/c/HaikoHertes

Haiko Hertes

Cloud Architect / Principal Consultant



Agenda

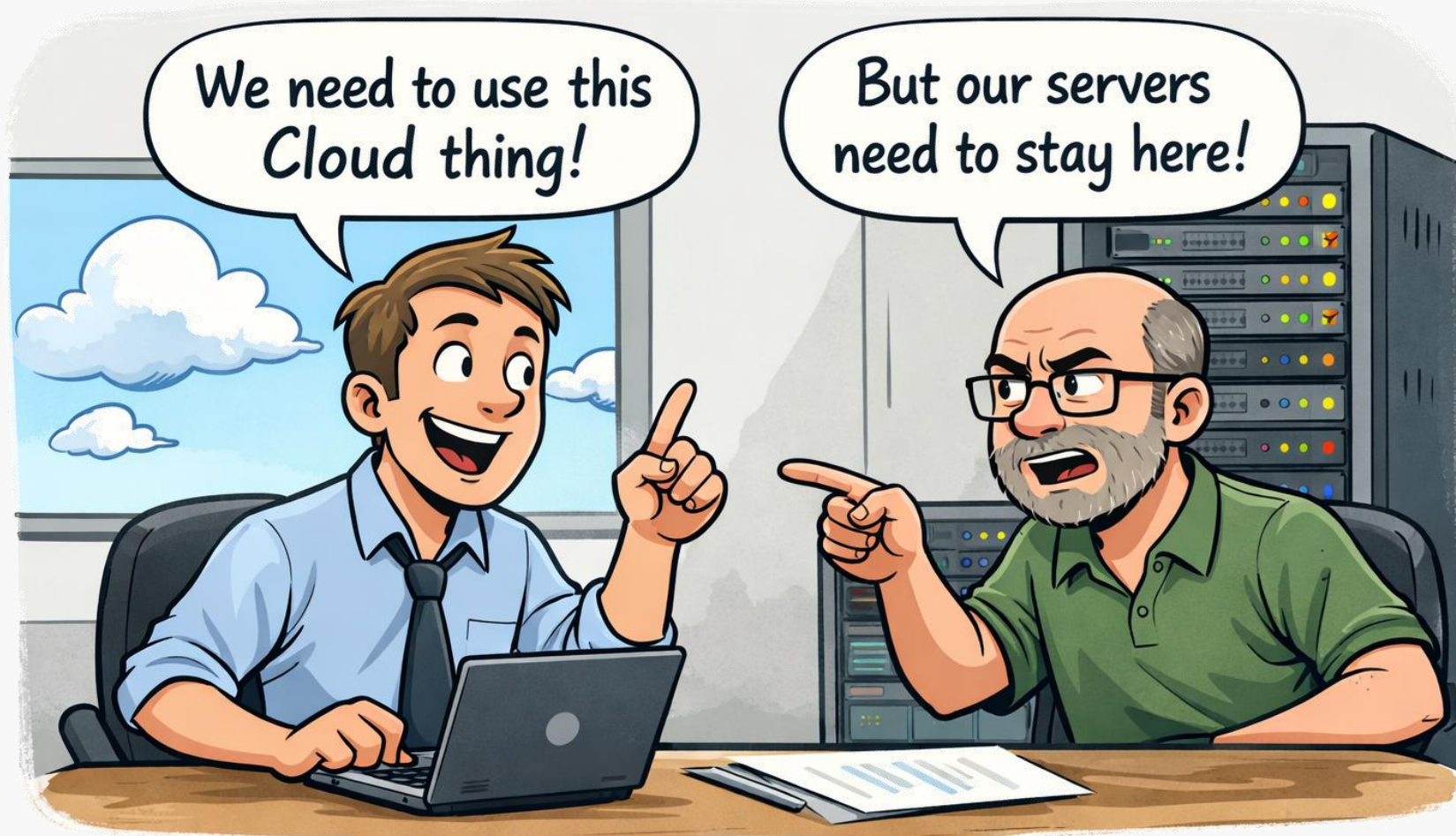
- Introduction
- What Azure Arc actually is
- Managing vCenter and SCVMM
- Day-2 operations
- Lessons learned

The typical datacenter

- Hypervisors and physical machines
- Multiple management tools
- Patching via scripts / WSUS / manual processes
- Monitoring split across systems
- Compliance verified locally
- No deep insights

=> *Stable, but fragmented!*

Cloud could be a solution....



*Most organizations are not choosing between cloud and on-prem -
they are negotiating between them!*

DEMO 01

**Let's jump right
into the very first
demo...**



What we have achieved

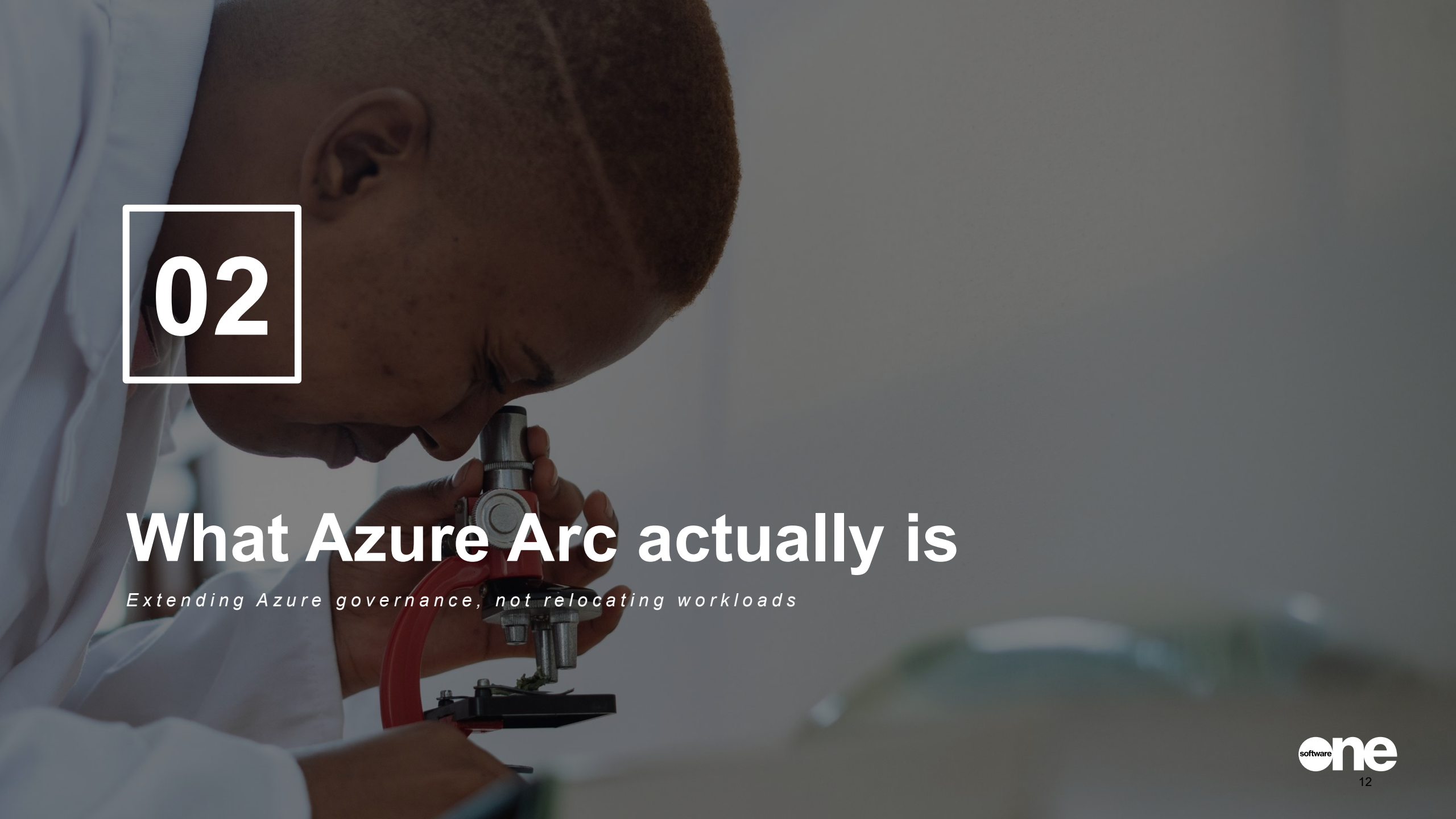
After onboarding, this on-prem server becomes:

- An Azure resource
- Governed by RBAC
- Configured via Policy
- Visible in centralized monitoring
- Without migration!

No cloud migration — only cloud control plane!

No inbound network connectivity required!



A person in a white lab coat is looking through a red microscope. The background is a soft, out-of-focus laboratory setting. A large white square with the number '02' is overlaid on the left side of the image.

02

What Azure Arc actually is

Extending Azure governance, not relocating workloads

IT CONNECTS RESOURCES TO AN API, NOT NETWORKS TO NETWORKS

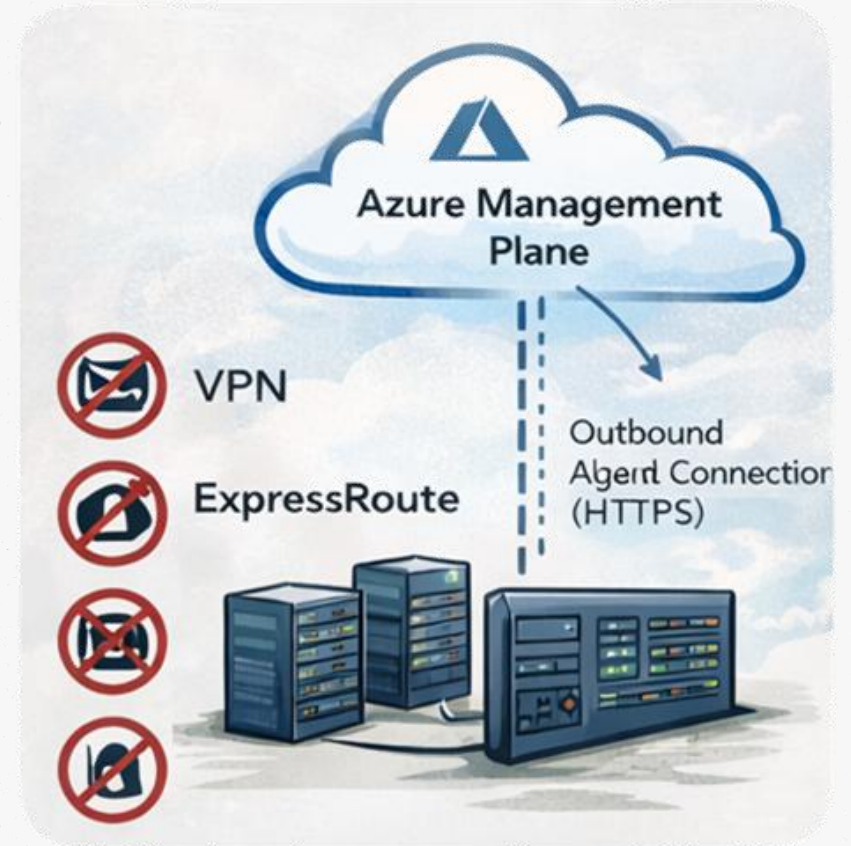
Azure Arc Is Not Hybrid Networking

Azure Arc does **not** require:

- VPN
- ExpressRoute
- Domain trust
- Inbound connectivity
- Moving workloads

Instead it provides:

- Azure management plane extended to external (any) machines



YOUR SERVER STAYS — ONLY CONTROL MOVES

The Key Concept

Data / Workload plane

- VM / workload runs where it already lives
- executes locally
- keeps its IP and dependencies

Control plane (Azure)

- ARM resource
- policy evaluation
- monitoring & automation
- identity & RBAC

Only management metadata flows to Azure!



Arc Connectivity Model

- Lightweight **agent** installed locally
- Outbound **HTTPS** (TCP 443)
- Authenticates via **Entra ID**
- Gets a **Managed Identity**
- Represents machine inside Azure Resource Manager

The machine becomes addressable without being reachable!

What Can Become an Azure Resource

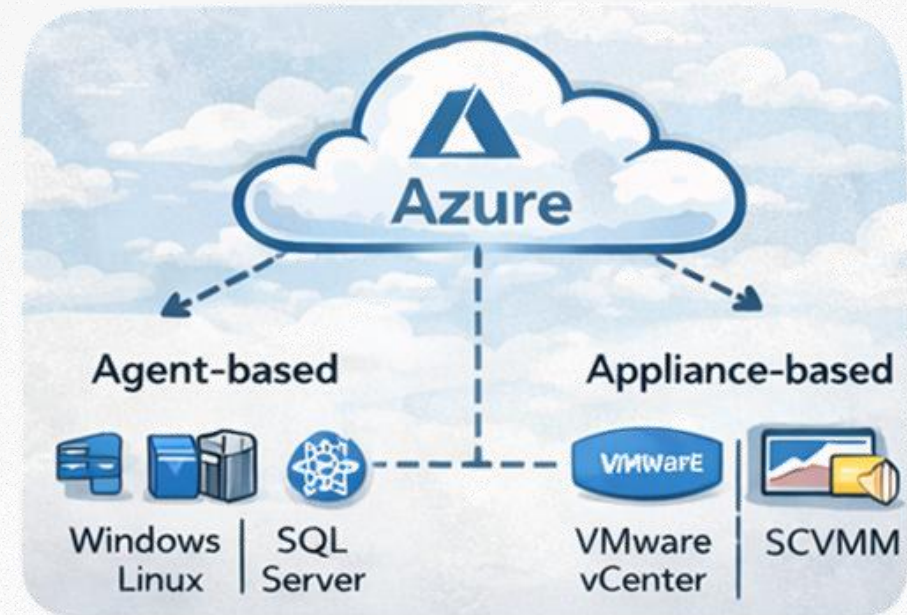
Agent-based

- Windows / Linux servers
- MS SQL Server
- Kubernetes

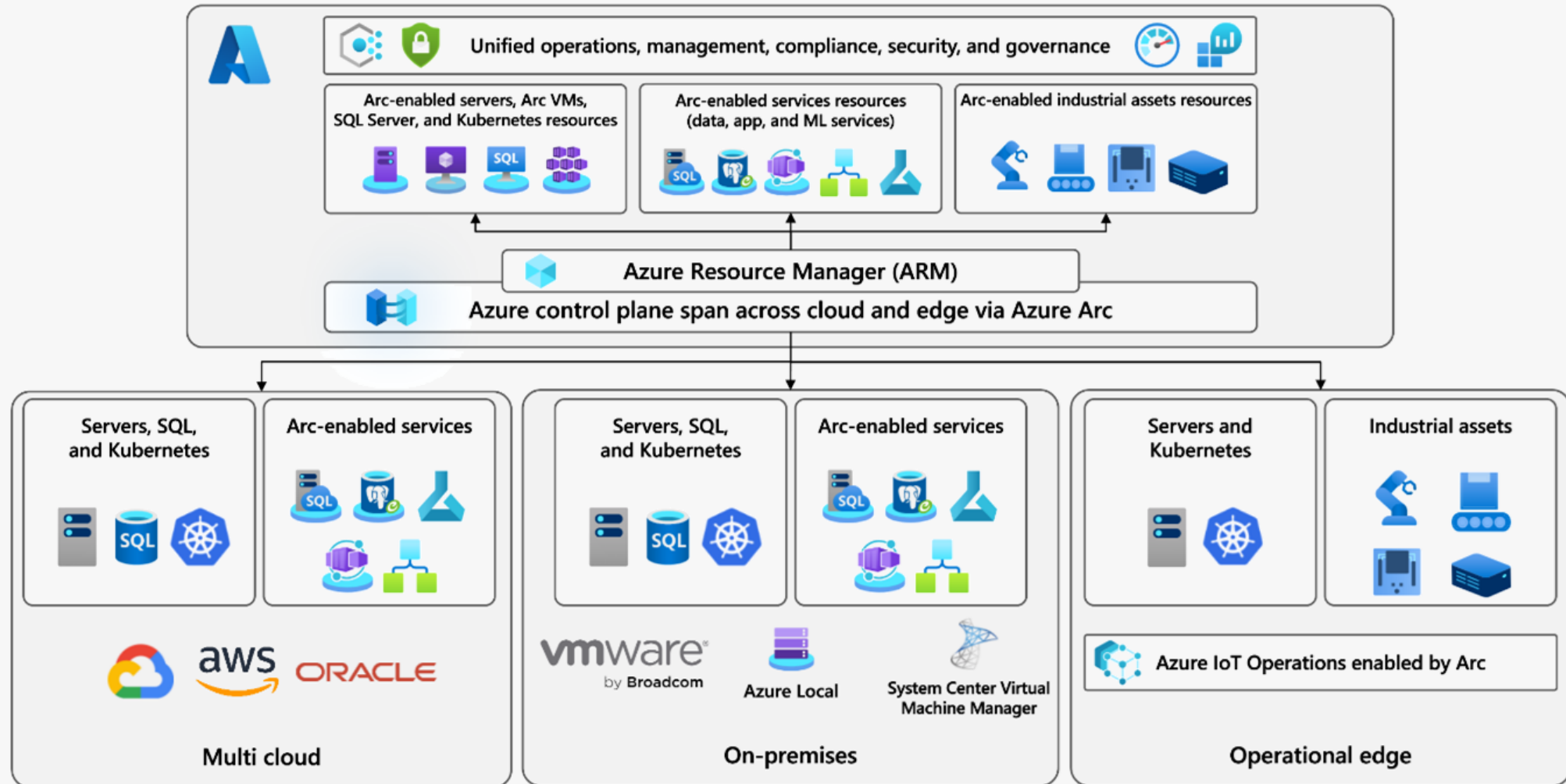
Appliance-based

- VMware vCenter
- SCVMM

Arc standardizes management across platforms!



What Can Become an Azure Resource



AND HOW MUCH IS THE FISH?

Arc Pricing

Azure Arc itself is free of charge (initially).

Azure Arc – core control plane

Services	Price	Notes
Inventory. Tag your resources, organize them into resource groups, subscriptions, and management groups, and query at scale with Azure Resource Graph to unify your environments.	FREE	N/A
Manage. Administrate your servers anywhere using SSH Arc, Run Command, and Custom Script Extension.	FREE	N/A
VM Self-service. Perform lifecycle management such as (create, resize, update and delete) and powercycle operations such as (start, stop, and restart on VMware vCenter and System Center Virtual Machine Manager Virtual Machines.	FREE	N/A

AND HOW MUCH IS THE FISH?

Arc Pricing

Management and Security Service for Azure Arc-enabled servers	Monthly Rate	Hourly Rate
Microsoft Defender for Servers Plan 1	\$5/server/month	\$0.007/server/hour
Microsoft Defender for Servers Plan 2	\$15/server/month	\$0.02/server/hour
Azure Update Manager	\$5/server/month	\$0.007/server/hour
Azure Policy Guest Configuration and Change Tracking & Inventory	\$6/server/month	\$0.009/server/hour
Hotpatching	\$1.50/core/month	\$0.00205/core/hour

These services are billed on a per GB ingested basis.

Management and Security Service for Azure Arc-enabled servers	Rate
Azure Monitor Analytics Logs	\$2.76/GB
Azure Monitor SCOM Managed Instance	\$6/month
Microsoft Sentinel	\$5.22/GB-ingested

AND HOW MUCH IS THE FISH?

Arc Pricing

There is several licensing options in Azure Arc, where you could pay for on-premises Licenses in a Pay-as-ou-go-model:

3

Windows Server Pay-As-You-Go (via Azure Arc)

Alternative to perpetual licensing.

Rate	Price
Per Core / Month	\$33.58
Per Core / Hour	\$0.046

Flexible activation & deactivation via Azure subscription.

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SQL Server Pay-As-You-Go (2012+)**

For customers without existing license:

Edition	Monthly	Hourly
Standard	\$73	\$0.10
Enterprise	\$273.75	\$0.375

5

Extended Security Updates (On-Prem / Multicloud via Arc)

For legacy workloads:

Windows Server 2012/R2 (per month)

Cores	Datacenter	Standard
16 Core	\$437	\$76
8 Core	\$219	\$38
2 Core	\$55	\$9.47

SQL Server

Version	Enterprise (2 Core)	Standard (2 Core)
SQL 2012	\$979	\$249
SQL 2014	\$1,081	\$278



03

Arc for Virtualization Platforms

From machine management to datacenter visibility

MANAGING SERVERS INDIVIDUALLY DOES NOT SCALE

Why Virtualization Integration Matters

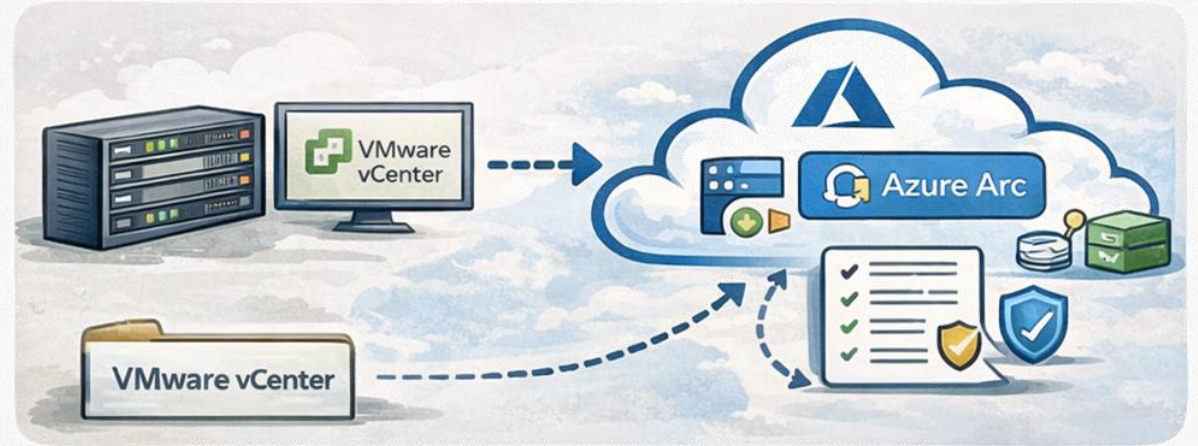
Without platform integration:

- VM lifecycle unknown to Azure
- Shadow IT VMs appear
- inconsistent tagging
- no governance before OS boot

With platform integration:

- Inventory before agent installation
- Governance at VM creation
- Datacenter visibility in Azure
- CRUD capabilities

Arc moves governance earlier in the lifecycle!



DISCOVER FIRST, ONBOARD LATER

VMware vCenter Integration

Azure connects to vCenter via Arc appliance

Provides:

- automatic VM discovery
- inventory in Azure Resource Graph
- VM state & metadata
- optional guest management onboarding

Key idea:

You see machines before you manage them!



SCVMM Integration

Same concept - different ecosystem:

- connects to SCVMM
- discovers Hyper-V VMs
- central inventory
- Arc onboarding from Azure

Important for:
Windows-heavy datacenters



A man and a woman are looking at a tablet together at night. The man is holding the tablet and pointing at the screen, while the woman looks on. They are both smiling. The background is dark with some blurred lights.

DEMO 02

VMWare vCenter & SCVMM Integration

Architecture Difference

Agent-based (servers):

- installed inside OS
- per-machine onboarding

Platform-based (vCenter/SCVMM):

- appliance connector
- inventory-first approach

*Arc can model infrastructure **before** touching workloads!*



A woman with dark hair in a ponytail, wearing a light pink button-down shirt, is looking down at a tablet computer she is holding. She is standing in a server room, with rows of server racks visible in the background. The racks have blue and green indicator lights. The overall lighting is dim, with the primary light source being the server lights.

04

Day-2 Operations with Azure Arc

From visibility to enforceable governance

VISIBILITY ALONE DOES NOT REDUCE RISK

The Shift from Inventory to Control

After onboarding, Azure Arc enables:

- Governance enforcement
- Patch orchestration
- Monitoring integration
- Security baseline control



Arc is not (only) inventory - it is operational leverage!

Azure Policy for Arc Servers

Examples:

- Enforce monitoring agent
- Require specific tags
- Enforce Defender
- Audit missing configuration

Policy evaluates machines (almost) like native Azure VMs!



Central / Unified Inventory

- All servers visible in Azure Resource Graph
- Queryable via KQL
- Cross-environment reporting
- Lifecycle tracking

Operational value:

You finally know what exists!

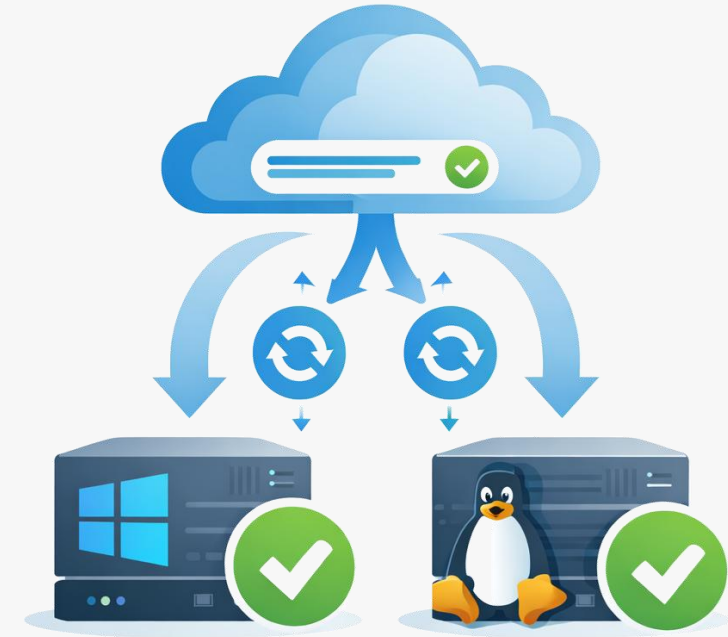
Modern Multi-OS Patch Management

Using Azure Update Manager:

- Assessment at scale
- Maintenance windows
- Staged rollout
- Cross-OS orchestration

Important:

No separate WSUS + Linux tooling required!
Update Management just tells VMs what to do.



Hotpatching

Hotpatching for Windows Server enables:

- Security updates applied without reboot
- Reduced maintenance windows
- Improved SLA compliance

Strategic value:

Security without operational disruption



Monitoring & Alerts

- Log Analytics integration
- Unified alerting
- Cross-platform insights
- Defender for Cloud integration
- AMBA-usage
(Azure Monitor Baseline Alerts)

Security and operations now see the same data!



Extended Security Updates (ESU)

Arc enables ESU delivery for:

- Windows Server
- SQL Server

Impact:

- Avoid forced migrations
- Maintain compliance posture
- Reduce emergency upgrade pressure

Keep in mind: ESU is not free!

A man and a woman are looking at a tablet together at night. The man is holding the tablet and pointing at the screen, while the woman looks on. They are both smiling and appear to be in a collaborative work environment. The background is dark with some blurred lights, suggesting an outdoor or semi-outdoor setting at night.

DEMO 04

Day-2-Ops

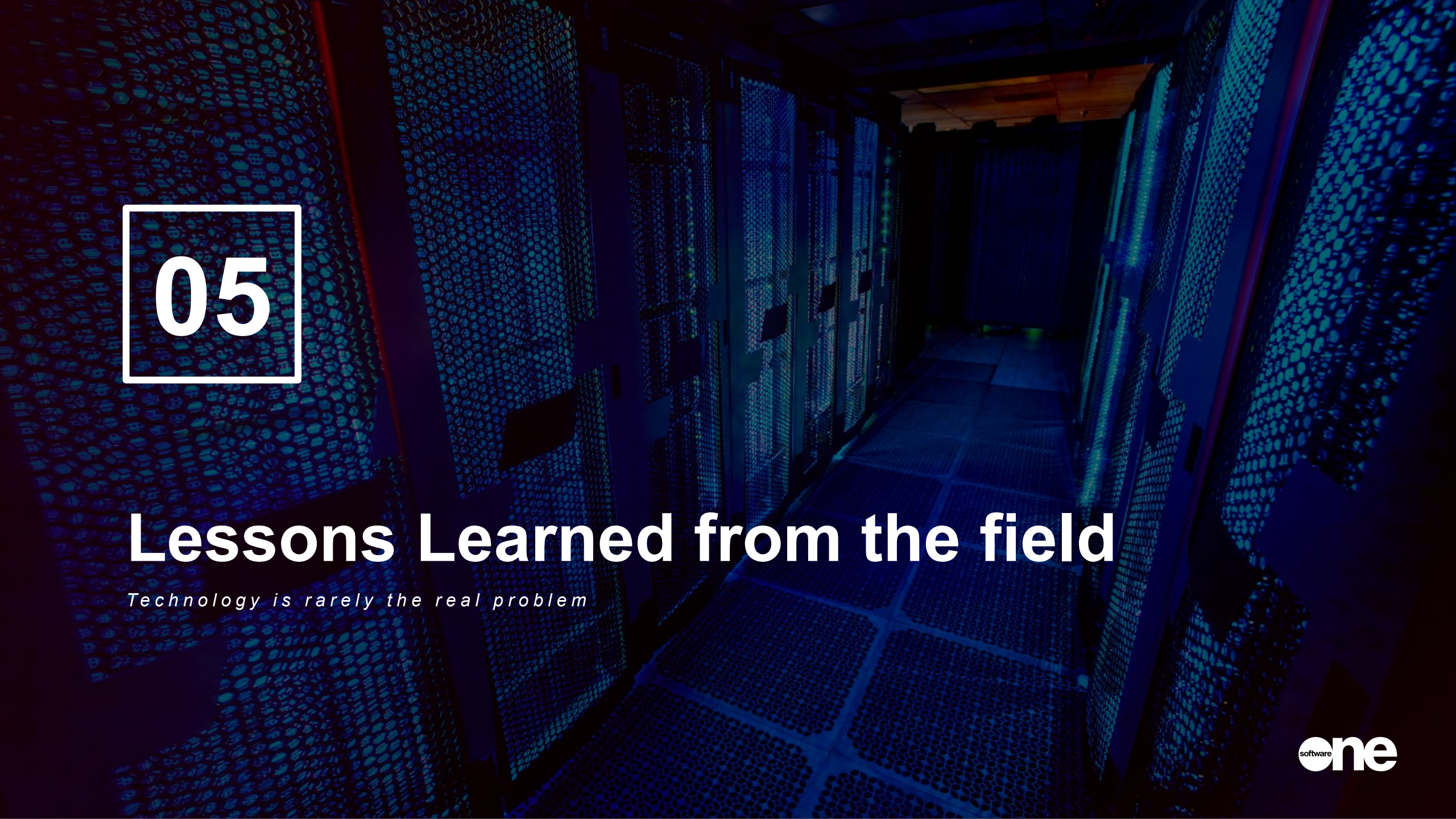
- Define Standards
- Orchestrate Operations
- Monitoring
- Security

What This Means Operationally

Azure Arc enables:

- Unified governance
- Consistent compliance
- Platform-wide reporting
- Gradual modernization

You modernize control before modernizing workloads.



05

Lessons Learned from the field

Technology is rarely the real problem

The Most Common Misconception

Common wrong expectation:

- *“Arc moves us to the cloud”*
- *“Arc replaces virtualization”*
- *“Arc eliminates legacy”*

Reality:

- Arc extends governance
- Workloads stay where they are
- It is a control-plane strategy

Arc is modernization of management - not modernization of workloads!

ARC AMPLIFIES WHATEVER STRUCTURE YOU ALREADY HAVE

No Governance Model = No Value

If you lack:

- clear subscription structure
- tagging standards
- RBAC model
- ownership model

=> *Then Arc will simply centralize the chaos.*

Technical impact:

- Policy sprawl
- inconsistent compliance results
- reporting noise

Design governance first - onboard second!

Underestimating Identity & RBAC

Frequent issues:

- Over-permissive roles
- No separation of duties
- No environment segmentation
- Shared accounts

Technical consequence:

- Excessive Owner assignments
- Audit complexity
- Security review friction

Arc is only as secure as your identity model!

Networking Assumptions

Common friction:

- Requests for VPN
- Concerns about inbound exposure
- SSL inspection breaking connectivity

Technical note:

- Outbound 443 required
- Time sync critical
- Use of Private Endpoints available!
- Proxy considerations must be planned

Security alignment early prevents rollout delays!

“ARC IS FREE” - PARTIALLY TRUE

Cost Expectations

Arc registration: free

But:

- Log Analytics costs
- Defender plans
- Update Manager usage
- ESU licensing

Mistake: Budgeting only for onboarding - not for operations.



Adoption Strategy Matters

Recommended approach:

- Pilot environment
- Governance validation
- Controlled rollout by workload type
- Reporting & optimization



Arc succeeds when introduced as a platform capability - not a side project!

06

Final Positioning & Takeaways

Modernizing control before modernizing workloads

What Azure Arc Really Delivers

Azure Arc provides:

- Central / unified inventory
- Governance enforcement via Policy
- Modern multi-OS patch management
- Extended Security Updates (ESU) support
- Hotpatch capabilities
- Integrated monitoring & security posture

Arc reduces operational fragmentation!

What Azure Arc Does Not Do

Arc does not:

- Replace virtualization platforms
- Eliminate technical debt
- Automatically modernize legacy apps
- Remove the need for architecture decisions

Arc improves management - not application architecture!

A Practical Maturity Model

1. Visibility
→ inventory & discovery
2. Governance
→ policy & tagging
3. Operations
→ patching & monitoring
4. Platform standardization
→ identity, security, lifecycle control

Most organizations stop at stage 2 – value increases significantly at stage 3!



HYBRID IS NOT TEMPORARY — IT IS OPERATIONAL REALITY

Final Thought

*Hybrid environments are not a transition phase.
They are a long-term operating model!*

Azure Arc provides:

- Control without relocation
- Standardization without disruption
- Modern governance without forced migration



**Before moving
workloads**

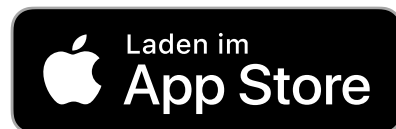
-

**move your
control plane!**





Bitte gebt uns euer Feedback in der App!



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